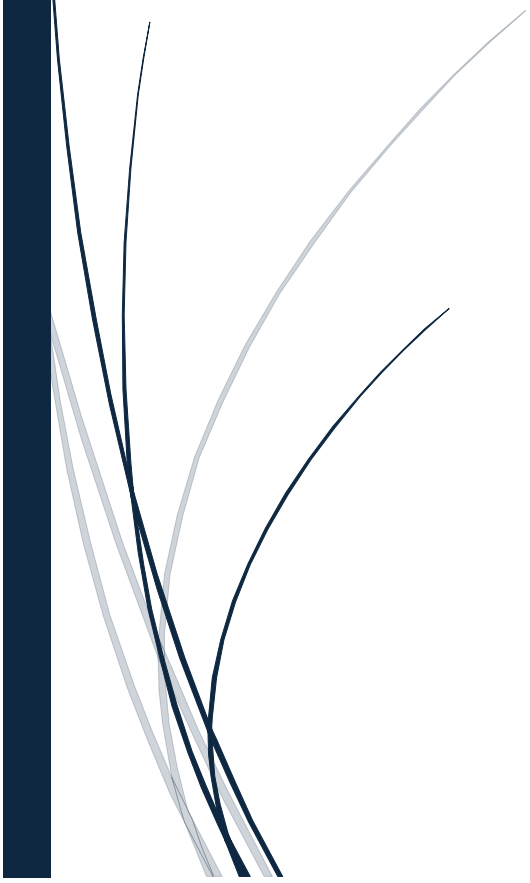
Assignment 14

Introduction to IP Addressing Theory



1. **List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.**

IPv4	**IPv6**
1. Uses 32-bit addresses.	1. Uses 128-bit addresses.
2. Example: 192.168.1.10	2. Example: 2001:db8::1
3. Supports about 4.3 billion addresses.	3. Supports a much larger number of addresses.
4. Written in decimal numbers separated by dots (.).	4. Written in hexadecimal numbers separated by colons (:).
5. NAT is commonly used to share public IPv4 addresses.	5. NAT is generally not required because of its huge address space.
6. Supports broadcast communication.	6. Does not support broadcast; uses multicast instead.
7. Routers can fragment packets when needed.	7. Only the sending device can fragment packets.

2. **Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.**

❖ **Network: 192.168.10.0/24**

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. **Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.**

Step 1: Subnet mask

- **CIDR: /27**
- **Subnet mask: 255.255.255.224**
- **Borrowed bits: (27-24=3\)**
- **Host bits remaining: (32-27=5\)**

Step 2: Calculate subnets and hosts

- **Number of subnets: (2^3 = 8\)**
- **Total IP addresses per subnet: (2^5 = 32\)**
- **Usable hosts per subnet: (32-2 = 30\)**

Block size = (256-224=32)

Subnet	Network address	Usable host IPs	Broadcast address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

4. **You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.**

Given:

- **IP address: 10.0.5.17**
- **Subnet mask: 255.255.255.0**

Step 1: Convert subnet mask to CIDR

255.255.255.0 = /24

Step 2: Calculate network address

Network address: 10.0.5.0

Step 3: Calculate broadcast address

Set all host bits to 1.

Broadcast address: 10.0.5.255

Step 4: Find valid host range

Exclude the network and broadcast addresses.

- First valid host: 10.0.5.1
- Last valid host: 10.0.5.254

Item	**Result**
IP address	10.0.5.17
CIDR notation	/24
Network address	10.0.5.0
Broadcast address	10.0.5.255
Valid host range	10.0.5.1 – 10.0.5.254
Usable hosts	254

5. **Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output**

IPv4 Subnet Calculator

Result

save

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	00001010000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

- **My manual calculation matched the online subnet calculator, and both gave the same network address, broadcast address, and valid host range.**